

Sameer Mohammad

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Professional Summary

Health Informatics professional with expertise in data analytics, clinical informatics, and healthcare software development. Skilled in ETL pipelines, ML dashboards, data visualization, and interoperability using Python, SQL, R, Power BI, Snowflake, and DHIS2. Proven experience across healthcare clients, academic research, and system development, delivering FHIR/HL7-compliant solutions. Adept at leveraging AI/ML, cloud technologies, and scalable databases to translate health data into actionable insights for evidence-based decision-making.

Technical Skills

- **Programming & ML:** Python, SQL, R, TensorFlow, PyTorch, Scikit-learn, NumPy
- **Statistics & Data Science:** Hypothesis Testing, Regression, Forecasting, Clustering, ANOVA
- **Visualization:** Power BI, Tableau, DHIS2, RedCap, Seaborn, ggplot, Matplotlib
- **Databases:** MySQL, PostgreSQL, PhpMyAdmin, Snowflake, REDCap, Oracle
- **ETL & APIs:** Airflow, FHIR API, DHIS2 API, JSON, REST APIs
- **Cloud & DevOps:** Docker, Git, GitHub
- **Healthcare Standards:** FHIR, HL7, SNOMED CT, ICD-10, RxNorm, OHDSI ATLAS, OpenEMR
- **Additional Skills:** Healthcare Analytics, Predictive Analytics, Machine Learning Algorithms, Data Analysis, Data Manipulation, Advanced Analytics, Database Management Systems, Information Technology, Information Systems, Analytical Skills, Project Management, Communications, Data Modeling, Ad Hoc Analysis, Problem-Solving Skills, Communication Skills

Professional Experience

Data Analyst – Molina Healthcare

Remote – USA | June 2025 – Present

- Designed and deployed a **Snowflake-based data warehouse** integrating **claims, EHR, and pharmacy datasets**, enabling real-time **analytics** for **population health**.
- Automated **ETL pipelines (Python, Airflow, SQL)** to ingest, transform, and validate **structured/unstructured data**, reducing manual processing time by **40%**.
- Built **FHIR-compliant APIs** to enable seamless **interoperability** between Molina's **care management platform** and external **EHRs**, ensuring **HIPAA compliance**.
- Developed **LLM-powered tools (LangChain, GPT-4)** to extract insights from **physician notes, discharge summaries**, and unstructured **EMR data**, boosting **clinical reporting efficiency**.
- Implemented **predictive ML models** for **patient risk stratification (readmissions, medication adherence)** using **scikit-learn** and **PyTorch**.
- Partnered with **clinical and IT stakeholders** to design interactive **Power BI dashboards**, improving **executive decision-making** and **care quality reporting**.
- Conducted **data quality audits** and **integrity checks**, identifying anomalies and improving downstream **accuracy of performance metrics**.

Health Data Analyst – Indiana University

Indianapolis, IN | Jan 2025 – May 2025

- Developed an automated **data integration pipeline** across **REDCap, DHIS2, and Power BI**, enabling real-time public health data visualization and reporting.
- Connected **REDCap API** to **Power BI** using **M language (Advanced Editor)**, **Web. Contents** and **custom JSON payloads** for secure data extraction and refresh scheduling.
- Designed **FHIR-aligned** and **DHIS2-compliant payloads** for **trackedEntityInstances, enrollments, and events**, improving data interoperability and ETL efficiency.
- Transformed health survey data using **Python (pandas)** and **Excel**, created validated **Data Dictionaries**, and imported structured datasets via Redcap's Data Import Tool.
- Published datasets to **Power BI Service**, enabling **Quick Insights, Q&A Visuals, and Smart Narratives** for natural language analytics and automated storytelling.
- Configured a full **Power BI Desktop-to-Service deployment pipeline**, supporting **cloud dashboards, scheduled refresh, and interactive filtering** for stakeholders.
- Applied best practices in **metadata modeling, data governance, and analytics documentation** to ensure reproducibility, security, and consistency across projects.

System Developer – Randolph County Caring Community Partnership (RCCCP)

Indianapolis, IN | May 2024 – May 2025

- Led full-stack development of a **DHIS2–Power BI decision support dashboard** integrating **machine learning models, geospatial visualization, and real-time public health analytics**.

- Installed and configured **DHIS2 (v41.0.0)** stack with **Apache Tomcat 9**, **PostgreSQL 14**, **PostGIS**, and **OpenJDK 11** for a secure local health data environment.
- Built and deployed **ETL pipelines** using **Python**, automating ingestion, transformation, and validation of **self-sufficiency, COVID-19, and PHQ-9 survey datasets**.
- Developed and validated multiple **ML models** using **Scikit-learn**, **TensorFlow**, and **Stats models**. Designed **interactive Power BI dashboards** with embedded Python scripts, geospatial maps, and Q&A-based insights; enabled live updates through **DHIS2 API integration**.
- Authored JSON payloads for **trackedEntityInstances**, **events**, and **enrollments** to ensure **FHIR-aligned data exchange** and cross-platform interoperability.
- Optimized **machine learning pipelines** through **data cleaning**, **feature scaling**, and **model tuning** ($R^2 = 0.62$, $F1 = 0.72$); developed **time-series models** (ARIMA, SARIMAX) for **forecasting**, integrated **predictive dashboards** for **RCCCP**, and conducted **usability testing** to ensure **data accuracy**, **scalability**, and **public health decision support**.

Data Science Intern - GTECHNOLOGIES PTY LTD

Indianapolis, IN | May 2024 – Aug 2024

- Designed and deployed an **AI-enabled EMR system** using **React**, **Node.js**, **MySQL**, and **Docker**, deployed on **Microsoft Azure**.
- Developed full-stack web applications, including a dynamic **patient registration form** with **auto-increment IDs**, **CSV data import**, and seamless front-end-backend integration.
- Built and normalized **relational databases** in **MySQL Workbench** and **DBeaver**; wrote and debugged complex **SQL queries**, **views**, and **stored procedures**.
- Resolved server-client architecture issues and integrated **Next.js** routing to improve **port configuration**, **scalability**, and **deployment stability**.
- Explored **Epic** and **Cerner** systems to understand **EMR interoperability**, aligning data workflows with **FHIR standards** and **structured clinical data flow**.
- Gained hands-on experience in **system deployment**, **cloud integration**, **code debugging**, and real-time **healthcare data processing** in production-grade environments.

Research Assistant- Biomedical Health Informatics Research Center (BHIRC), IU

Indianapolis, IN | Jan 2024 – May 2024

- Improved **EHR system performance** by optimizing **data workflows** and enhancing **data quality** for **clinical research**. Collaborated with **faculty** and **technical teams** to support **analytics pipelines**, **validate data outputs**, and contribute to **healthcare process improvement** through **informatics-driven insights**.
- Researched and compiled **datasets** related to **Children's and Family Health**, including data from **CMS**, **CDC**, **HRSA**, and **SAMHSA**. Investigated **government policies**, **clinical capabilities**, **academic infrastructure**, and **population characteristics**, transforming findings into **structured database entries**. Designed and **normalized a MySQL database** to organize **source metadata**; performed **data wrangling** and formatting in **Excel** and **Python (pandas)** for ingestion.
- Strengthened skills in **federal health data sourcing**, **research data curation**, and **informatics-driven knowledge organization** to support ongoing **faculty and student research initiatives**.

Clinical Data Analyst – ESIC Hospital, MLRIP.

Hyderabad, IN | June 2022 – July 2023

- Assisted in ward rounds across multiple specialties, identifying **ADRs** and ensuring accurate documentation. Led a 5-member team on **clinical research** projects, overseeing data collection and **statistical analysis**, and conducting patient history reviews to support diagnosis and treatment planning.
- Performed statistical tests for a breastfeeding study among primi mothers, collaborating with multidisciplinary teams to optimize clinical workflows and **patient care**.

Academic Projects

- Esophageal Cancer Analytics System – MySQL schema + ML models for treatment outcomes
- AI-Enabled EMR System Development– Full-stack EMR with React, Node.js, MySQL.
- Global Food Security & Climate Change – R-based regression, clustering, and geospatial analysis
- Lung Cancer Predictive Model – Python-based ML risk model
- Remote Insulin Device Evaluation – Clinical usability testing and adoption strategy

Education

Indiana University – Indianapolis, IN

M.S. Health Informatics (GPA: 3.89) | Aug 2023 – May 2025

Marri Laxman Reddy Institute of Pharmacy – Hyderabad, India

Doctor of Pharmacy (Pharm.D) (GPA: 3.8) | Jun 2017 – Jun 2023